Daerhan Jianati

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Education

University of Illinois Urbana-Champaign

Aug. 2026–May 2028

School of Information Sciences
M.S. in Information Management

Nanjing University

Sep. 2022–Jun. 2026

School of Electronic Science and Engineering
B.S. in Electronic Information Science & Technology
Honor: Ethnic Minority Students' Pomegranate Seed Scholarship, 2024 and 2025

Research Interests

Computational Imaging | Bio-Inspired Visual Perception | Multimodal Sensor Fusion
Robot Vision | Spectral Imaging | Human Visual Perception

Selected Research Experience

Bio-Inspired Multi-Agent Perception for RGB–NIR Robot Vision

Sep. 2026–Present

Ongoing Research; multimodal visual perception and geometric estimation

- Developing an insect-inspired perception framework extending prior RGB–active NIR fusion into lightweight sensing and task-specialized modules for low-light, motion, depth, and visual memory.
- Designed structured shared perceptual memory with on-demand cross-agent retrieval; prototyped feedback between depth estimation and visual pose/memory modules.
- Evaluating depth, visual odometry/pose, feature matching, temporal consistency, and computational cost, with optional sparse LiDAR as an additional geometric input.

Geometry-Guided Diffusion Restoration for RGB–NIR Imaging

2026–Present

Ongoing Research; low-light restoration and geometric consistency

- Developing region-aware diffusion restoration for simultaneous RGB underexposure and active-NIR saturation, building on failure cases identified during prior fusion research.
- Exploring geometry/depth guidance and spatial confidence masks; designing evaluations for depth, feature matching, and temporal consistency. Experiments are ongoing.

Self-Supervised RGB–Active NIR Fusion for Geometric Perception

Nov. 2025–May 2026

Completed Research; collaboration with Prof. Guoyu Lu, Indiana University

- Developed a lightweight single-channel RGB–NIR representation using pixel-wise fusion, stereo reprojection, and an exposure-aware reliability gate.
- Evaluated the representation on stereo depth, feature matching, SfM/SLAM diagnostics, detection, and efficiency (79.36 FPS; 560 MB peak memory in the evaluated configuration).
- This completed work informs ongoing diffusion restoration and bio-inspired multi-agent perception research. **Code**

Per-Pixel Illumination Estimation from HR-RGB and LR-HSI

Jun. 2025–Oct. 2025

Student Researcher, CITE Lab, Nanjing University

- Developed an RGB–hyperspectral sensing pipeline for per-pixel illumination estimation; reproduced baselines and tested spatial downsampling and fusion settings.

Unitree Go2 Multi-Floor Guide Navigation

Apr. 2026–Jul. 2026

Research Assistant, Human-Computer Interaction Lab, Nanjing University

- Built a ROS 2/Nav2 guide-robot prototype with LiDAR-based mapping, localization, global planning, motion

integration, semantic destinations, and a human-assisted elevator workflow.

- Open-sourced code, maps, configurations, and bilingual documentation. **Code** | **Docs**

Technical Skills

Programming	Python, C, MATLAB, Arduino
Imaging & Vision	RGB–NIR fusion, depth/pose estimation, feature matching, illumination estimation, geometric evaluation
Robotics	ROS 2, Nav2, visual odometry/SLAM, modular perception, Unitree Go2
Sensors & Data	RGB, active NIR, hyperspectral, polarization, LiDAR
Tools	PyTorch, OpenCV, NumPy, Linux, Git, LaTeX

Additional Research & Projects

Exploratory RGB/NIR/Polarization Depth Estimation

Aug. 2025–Oct. 2025

Explored multimodal fusion, custom losses, knowledge distillation, calibration, and sparse LiDAR supervision.

3D Dynamic Heart Reconstruction from 2D Medical Imaging

Feb. 2025–May 2025

Project Lead, Course Project; led preprocessing, segmentation, reconstruction, and myocardial-motion visualization.

Intelligent Risk Butler: Exchange-Rate Risk Inspection System

Jan. 2025–Jun. 2025

Core Member, Algorithm Group, 20th Citi Cup; developed an LLM module and exchange-rate risk signal pipeline.

Quantum Implementation of Finite Element Analysis

Jan. 2025–Feb. 2025

Exchange Project, North Carolina State University; implemented block-encoding/QSP-based solvers and explored BPX preconditioning.

Reservoir-Computing-Transistor-Based Fall Detection

Nov. 2023–Dec. 2024

National Undergraduate Innovation Program; contributed to a hardware-software system integrating analog computing, Arduino, Bluetooth, and vision-based motion analysis.

Industry Experience

Data Analysis Intern

Jul. 2024–Sep. 2024

College of Water Conservancy and Architectural Engineering, Shihezi University

- Processed multivariate environmental sensor data and evaluated machine-learning models for anomaly detection in irrigation leakage and drought stress.

Research Outputs

Unpublished Research Manuscript

2026

Daerhan Jianati and G. Lu. "Learning a Single-Channel Fusion Representation from RGB and Active NIR via Self-Supervision." (Unpublished.)

Patent Application

Jan. 2025

"A System and Method for Elderly Fall Detection Based on Reservoir Computing Transistors." Fifth inventor; Application No. CN202411352818.X.